UDROŻNIENIE ŁÓDZKIEGO WĘZŁA KOLEJOWEGO (TEN-T), ETAP II, ODCINEK ŁÓDŹ FABRYCZNA - ŁÓDŹ KALISKA/ŁÓDŹ ŻABIENIEC – PROJEKT POIIŚ 5.1-15

TBM 13.08 M - CONSTRUCTION FOLLOW-UP

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Client	PBDiM - Energopol
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1. INTRODUCTION

S-1222 TBM EPB machine with 13.08 m excavation diameter is currently mining Axis 23, in the section between Polesie Station and Śródmieście Station. TBM has restarted from chainage 1+962.

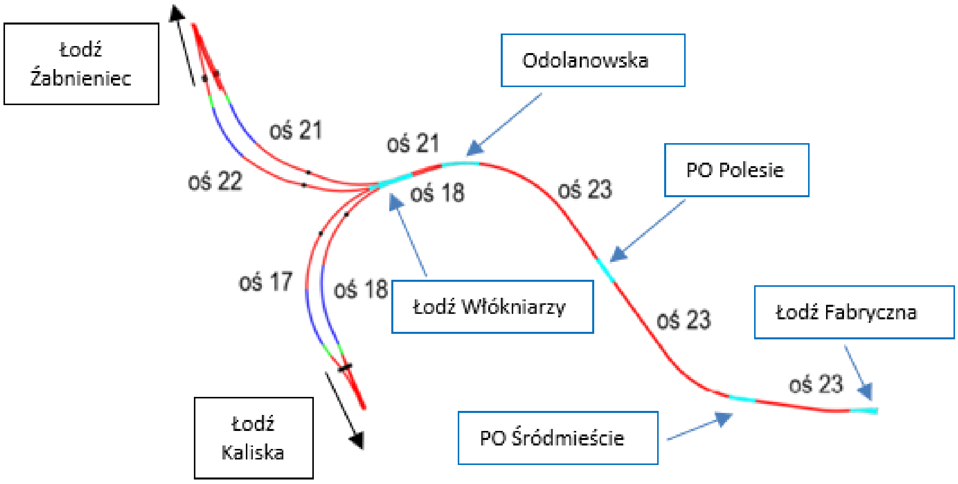

Figure 1-1 Overview of main structural elements of the Project

This daily follow-up report aims to give a general overview of S-1222 TBM drive during 29/08/2024. It includes EPB face pressure, belt scale parameters, thrust and torque values, primary grout parameters, guidance, as well as monitoring follow-up data.

According to VMT system, TBM is at Chainage 1+680.12 after the erection of ring n. 237; which corresponds to a total excavation length of 286.35 m from the starting chainage.

Figure 1-2 Plan view of TBM Position on 30/08/2024

2. FACE PRESSURES AND EXCAVATION VOLUMES

Average EPB pressure at the crown is at about 1.8 bar. The value is within the attention values. There is a drop of face pressure at the begin of excavation of ring 238.

Figure 2-1 shows EPB sensors.

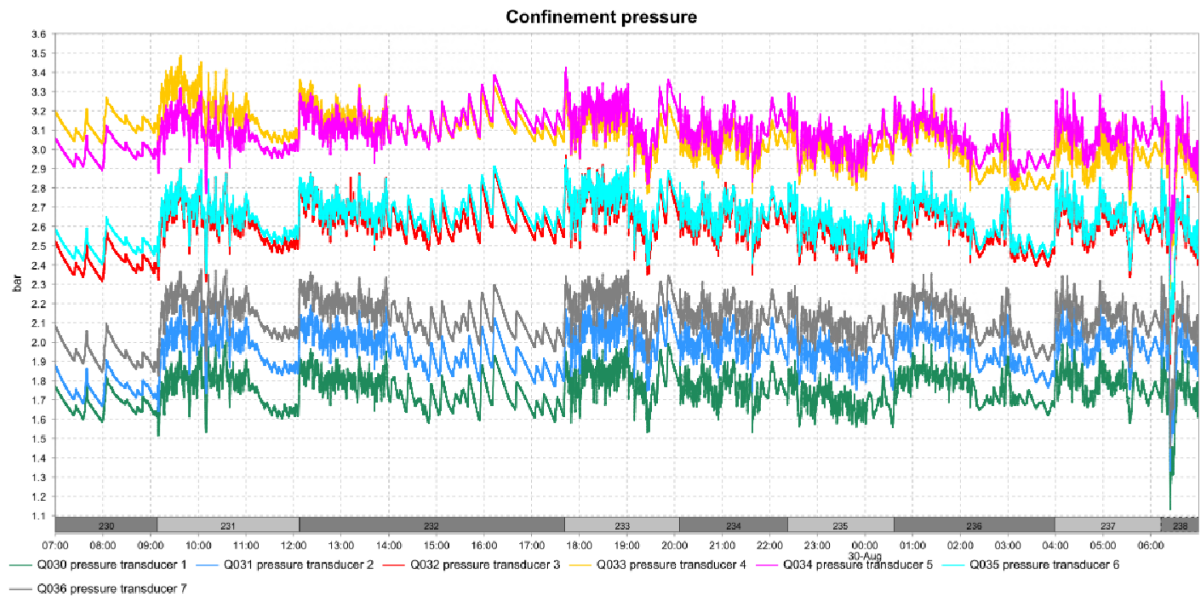

Figure 2-1 EPB pressure sensor

The delta between pressure transducer 1 and 2 has an average value of 0.2 bar indicating that the bulk chamber is correctly filled with material. During excavation of ring 230 the delta value is low (about 0.8 bar) indicating that the upper part of the chamber is not correctly filled with conditioned soil.

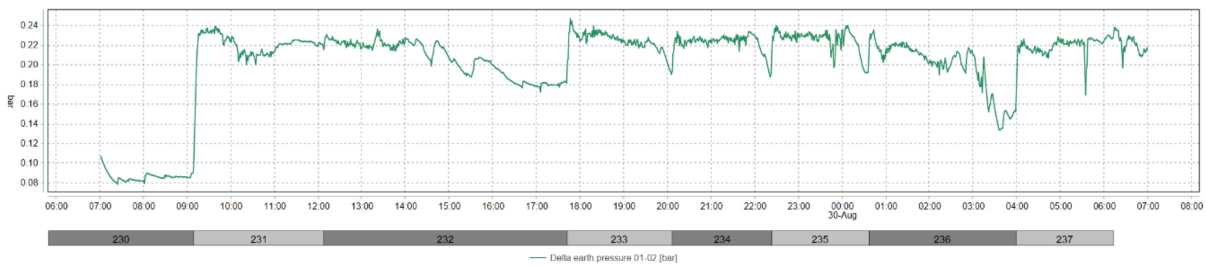

Figure 2-2 Delta between pressure sensors 1 and 2

Belt scale n. 2 show values much lower than n. 1. The results of Belt scale n. 2 seems to be affected by a constant error and are therefore considered not representative. The belt scale value is a raw indication of the excavated material quantity. Excavated material weight is higher than expected from the design in all rings.

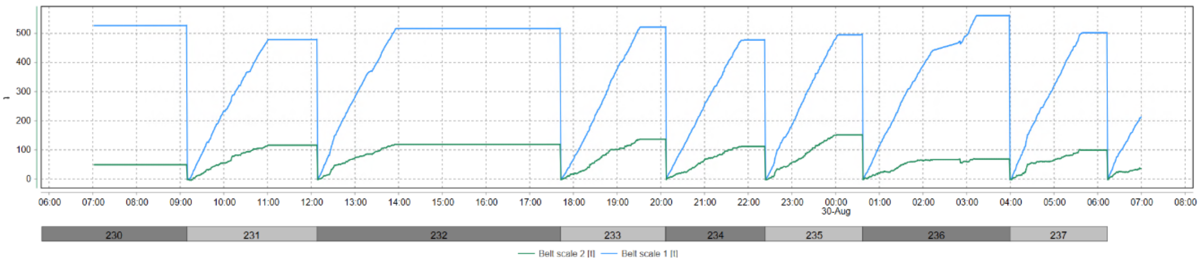

Figure 2-3 Belt scales

Table 2-1 Belt scale values at the end of each advance

Ring	Belt scale 1 (t)	Belt scale 2 (t)
230	528	51
231	480	118
232	518	120
233	523	138
234	478	114
235	498	153
236	562	71
237	503	102

3. THRUST AND CUTTERHEAD TORQUE

During excavation, the average torque is of 7.2 MNm.

During advancement, average advance rate is of 16 mm/min.

During excavation, thrust force shows an average value of 59.2 MN and a maximum of 74.7 MN.

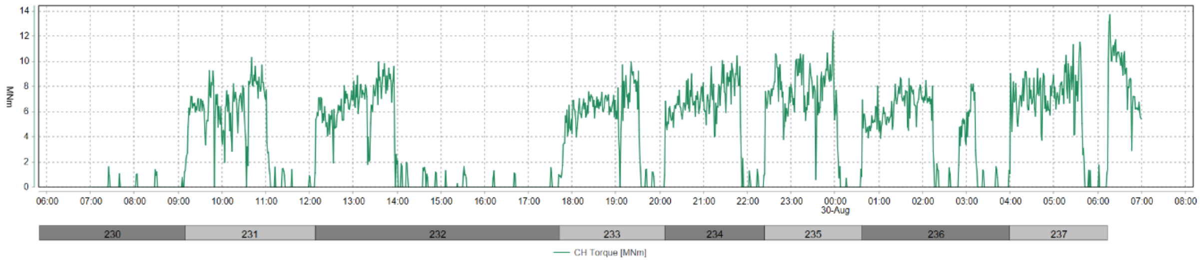

Figure 3-1 CH Torque

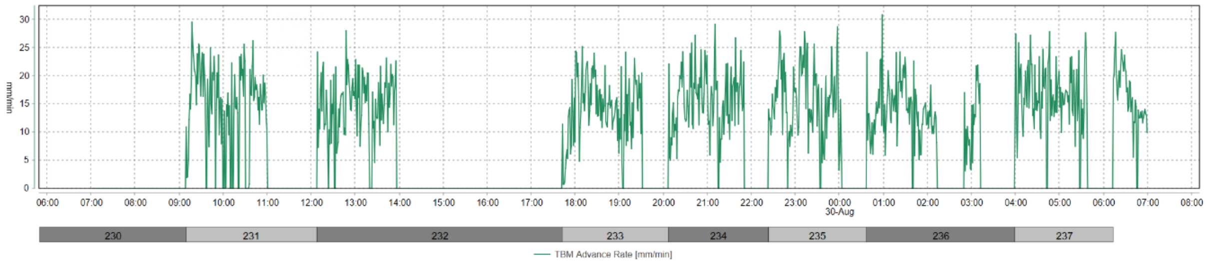

Figure 3-2 Advance rate

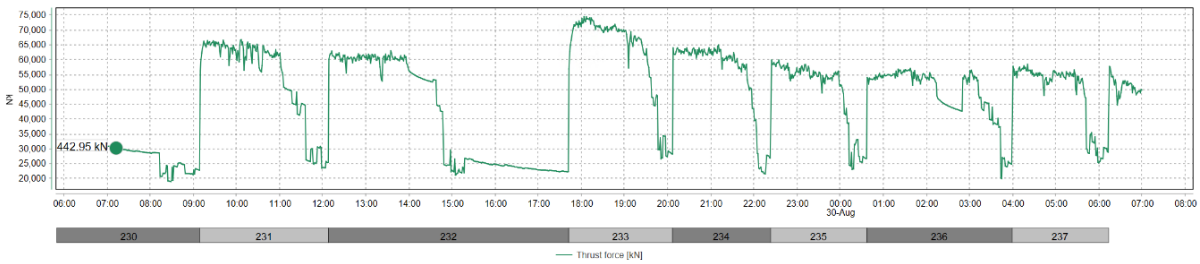

Figure 3-3 Thrust force

4. GROUT INJECTION

Grout injection has a volume above the minimum theoretical annular volume +%10 of 13.54 m^3, except for rings 232, 233 and 237 that have volumes higher than the theoretical one, but lower than 13.54 m3. Ration B/A is at 7%.

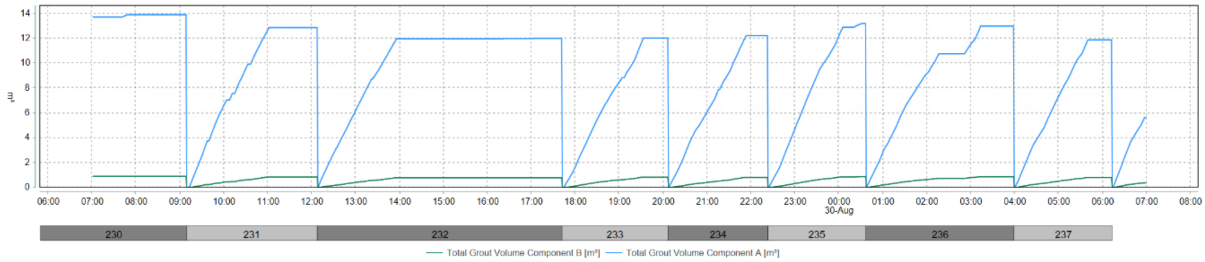

Figure 4-1 Grout volumes

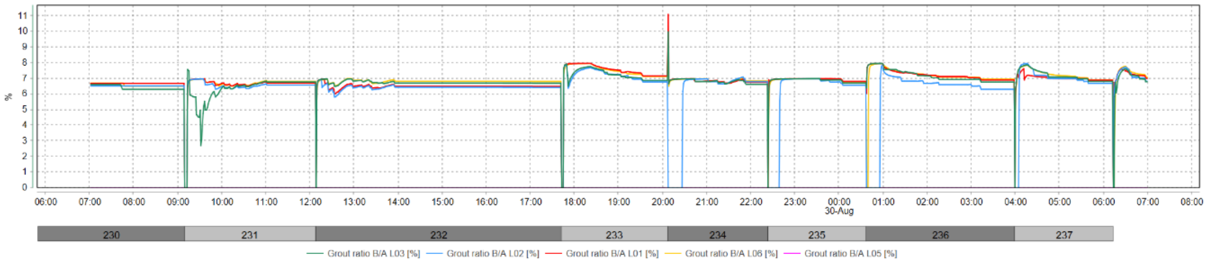

Figure 4-2 Volume ratio Components B/A

Table 4-1 Grout volumes per ring

Ring	Component A (m^3)	Component B (m^3)	B/A (%)
230	13.9	0.9	7
231	12.9	0.9	7
232	12	0.8	7
233	12	0.8	7
234	12.2	0.8	7
235	13.2	0.9	7
236	13	0.9	7
237	11.9	0.8	7

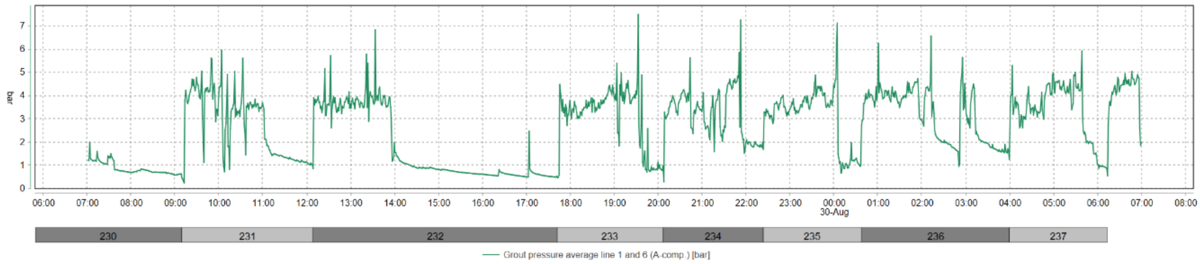

Figure 4-3 Average injection pressure line 1 and 6

5. GUIDANCE

The TBM vertical deviation is of about 13.58 mm on the front edge of the shield and the horizontal deviation of about -1.21 mm (Figure 5-1). Maximum radial deviation is of 15.24 mm.

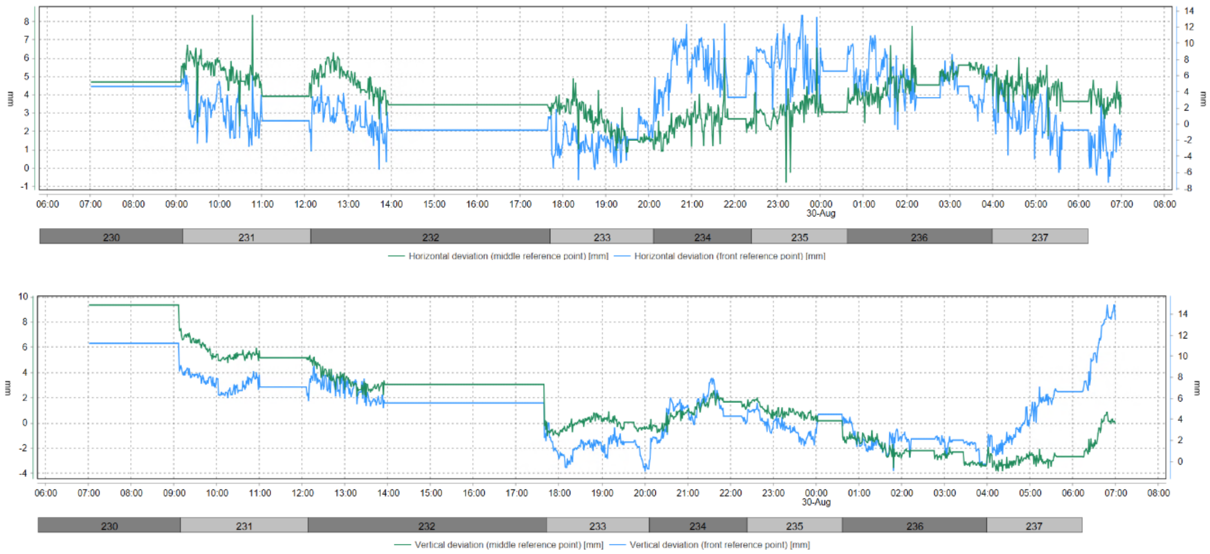

Figure 5-1 Machine deviation from the alignment

6. MONITORING

According to Geoscope, settlements of the buildings above the tunnel are kept within the limits.

Monitoring data are not updated during the night shift.

Figure 6-1 Monitoring points

The monitoring points on Prochnika street show settlement values of about 1 mm in 24 hours, particularly in point L23_308, L3_309, L23_314 and L23_315, that are the point closest to tunnel axis.

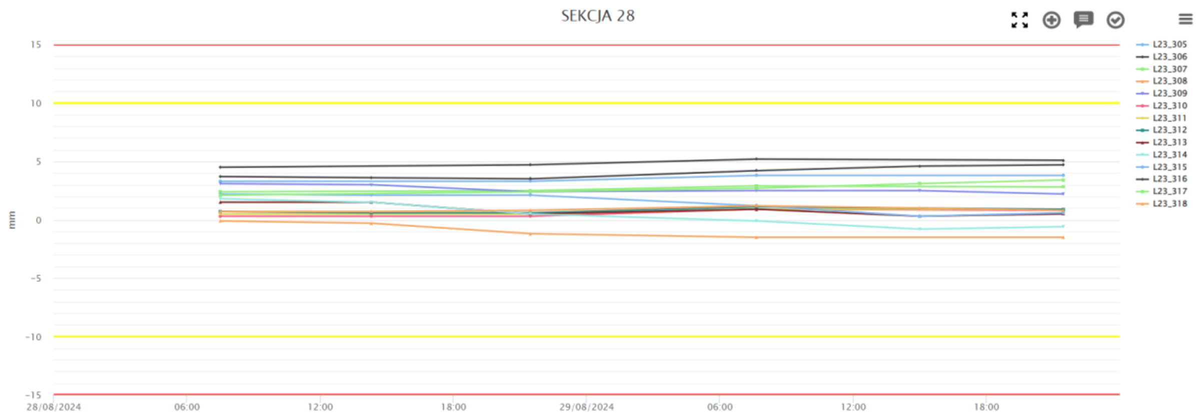

Figure 6-2 Settlement of Section on Prochnika street

On PR0450, monitoring points are stable.

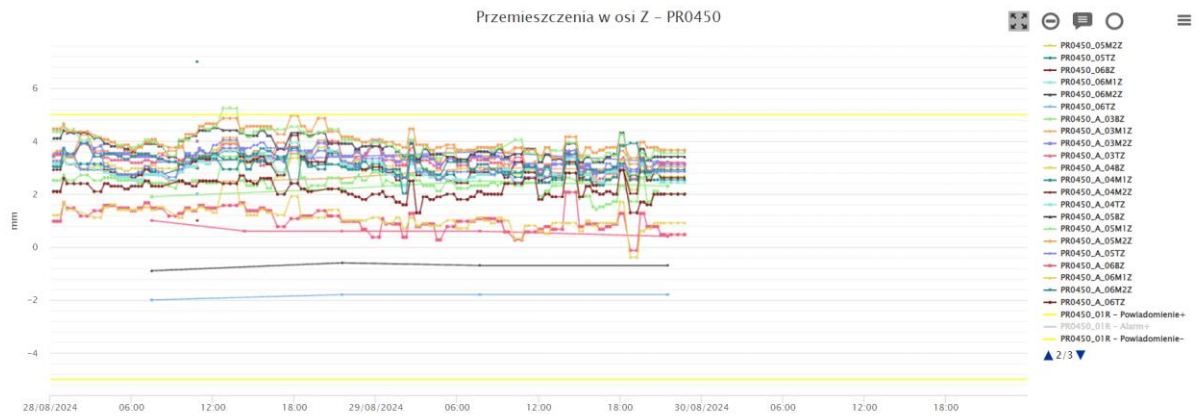

Figure 6-3 Settlement of building PR0450

On PR0460, monitoring points show a small downward trend. The absence of monitoring values after 22.00 does not permit to evaluate the trend development.

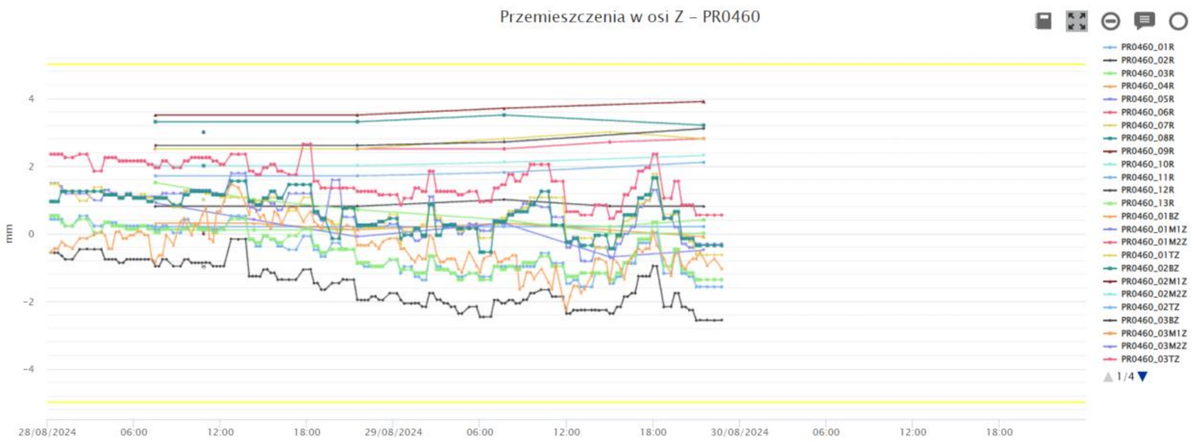

Figure 6-4 Settlement of building PR0460

On PR0440, monitoring points show a small upward trend. The absence of monitoring values after 22.00 does not permit to evaluate the trend development.

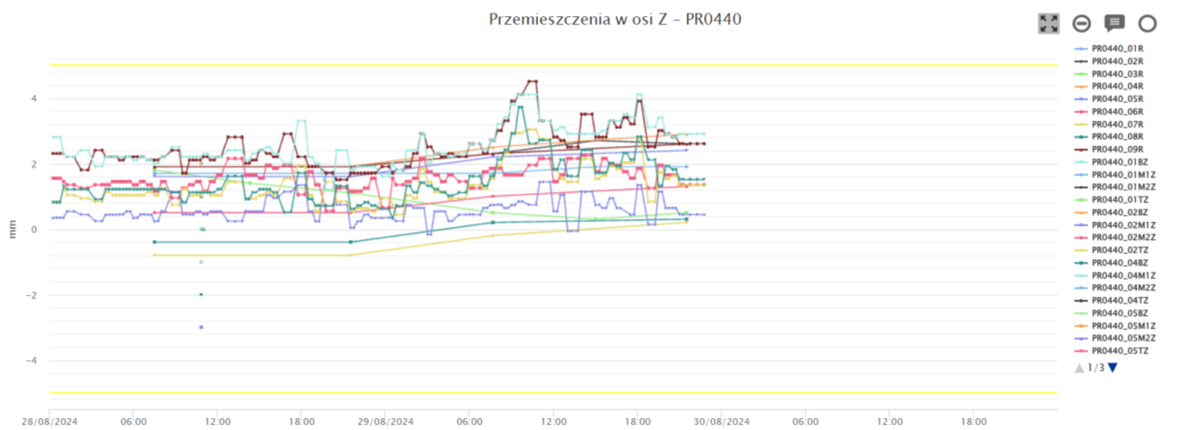

Figure 6-5 Settlement of building PR0440

7. COMPENSATION GROUTING ACTIVITIES

According to Raport Dobowy 29/08/2024 of Soletanche, compensation grouting has not been performed on the 29/08/2024 from Shaft 2 in Próchnika 45.

According to Raport Dobowy 29/08/2024 of Soletanche, compensation grouting has been performed on the 29/08/2024 from Shaft 3 in Próchnika 42.

The total amount of declared injected volume is 14995 l.

8. SOIL CONDITIONING

The volume of injected foam is plotted in the following Figure. The FIR value is computed based on the excavated material amount per ring and reported in the Table.

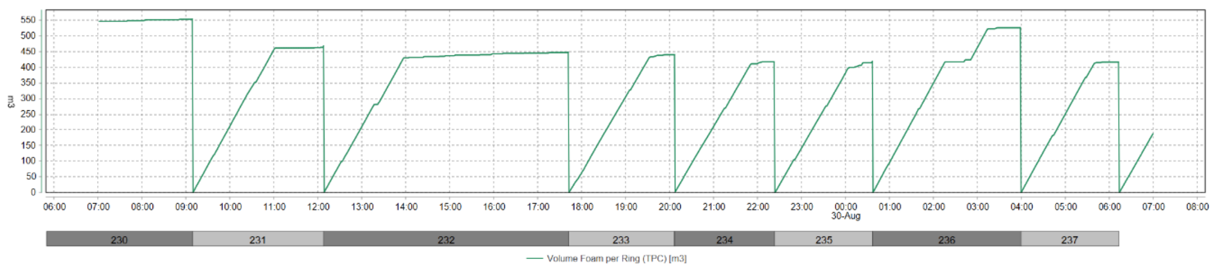

Figure 8-1 Injected foam volume

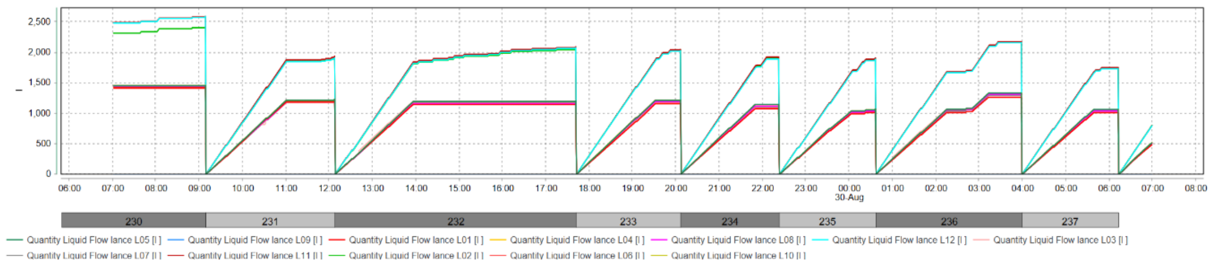

Figure 8-2 Conditioning liquid volume

Table 8-1 Soil conditioning data

Ring	Liquid volume (m^3)	C (%)	FIR (%)
230	19.2	2	261
231	15.5	2	221
232	15.8	2	212
233	15.8	2	208
234	14.7	2	197
235	14.2	2	198
236	17.1	2	248
237	13.6	2	197